06/08/26 MLA-0406 Deep Learning

CO2 CLASS-TEST

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Q1) What is Active function needed, what are all they?

A) * Here, Active function introduce to an low-linearity for the neural networks

* In this neural networks we are using complex patterns
* without complex patterns we are unable to active function for the neural networks.

Comparision Table :-

Functions	~~Formula~~
Sigmoid	→ It is a function mainly used to sort the mail is spam
Tanh	→ it is a function used for the analysis of the stock market trends
ReLU	→ It is used for classification of images
ELU	→
LeLU	→

* for an example for the neural network how does it works → In a college their is a lot of data Students & Staff's and management for each category their will be a different function's of neural networks work's.

2Q) Sigmoid, Tanh, ReLU Derivations & Gradient problems

A) Mathematical Expressions :-

* Sigmoid : $\sigma(x) = 1/(1+e^{-x})$
* Tanh : $\tanh(x) = (e^{x}-e^{-x})/(e^{x}+e^{-x})$
* ReLU : $f(x) = \max(0, x)$

Gradient in Backpropagation :-

Q3)

* Gradients calculate how much weight should change
* weights are updated using gradient.

-> vanishing Gradient :-
* Gradient becomes very small
* Training becomes slow
* occors mainly with sigmoid and Tanh.

-> Exploding Gradient :-
* Gradient becomes very large
* Training becomes unstable.
* solved using Relu, gradient

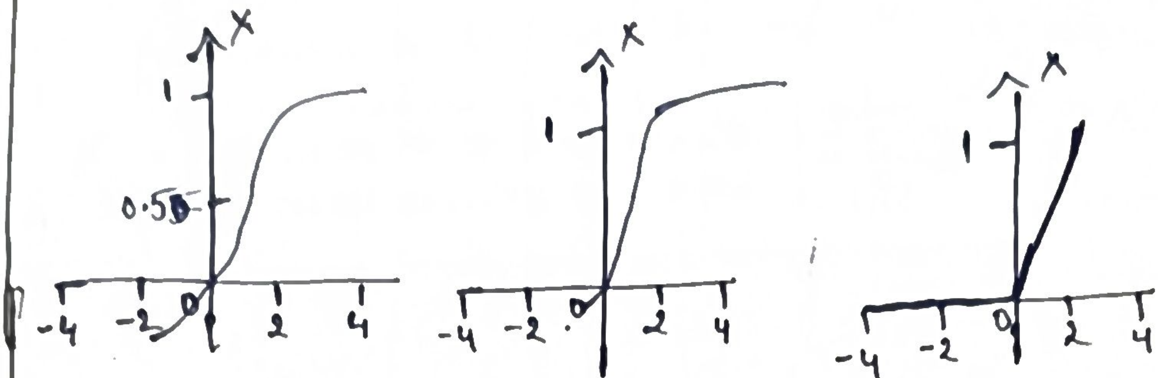
4Q) Effect of Activation fonctions ?

A) * it will transfer ReLU
* in this Activation function Sigmoid and Tanh are slower.
* But in the computational effeciency ReLU will be the fastest
* and also as well as sigmoid and Tanh require more computation.

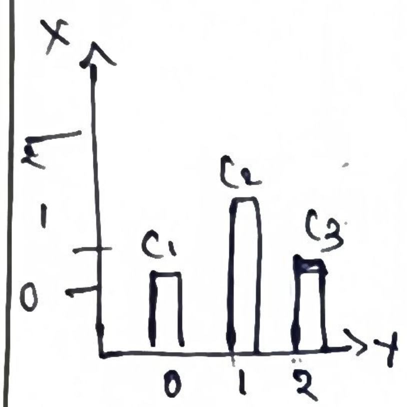
Examples :-
* Image classification -> ReLU
* House price prediction -> Sigmoid
* student prediction -> sigmoid

-> Here Relu is best for hidden Layers, sigmoid for binary classification.

Q3) Graphs :-



Sigmoid Tanh ReLU



ELU

Table :-

Image classifications	ReLU	~~Sol~~ Sigmoid
Binary classification	ReLU	Softmax
Regression	ReLU	linear (no active)

Justification :-

* ReLU -> fastest
* Sigmoid -> Binary output (0 - 1)
* softmax -> class probabilities
* Regression -> Linear